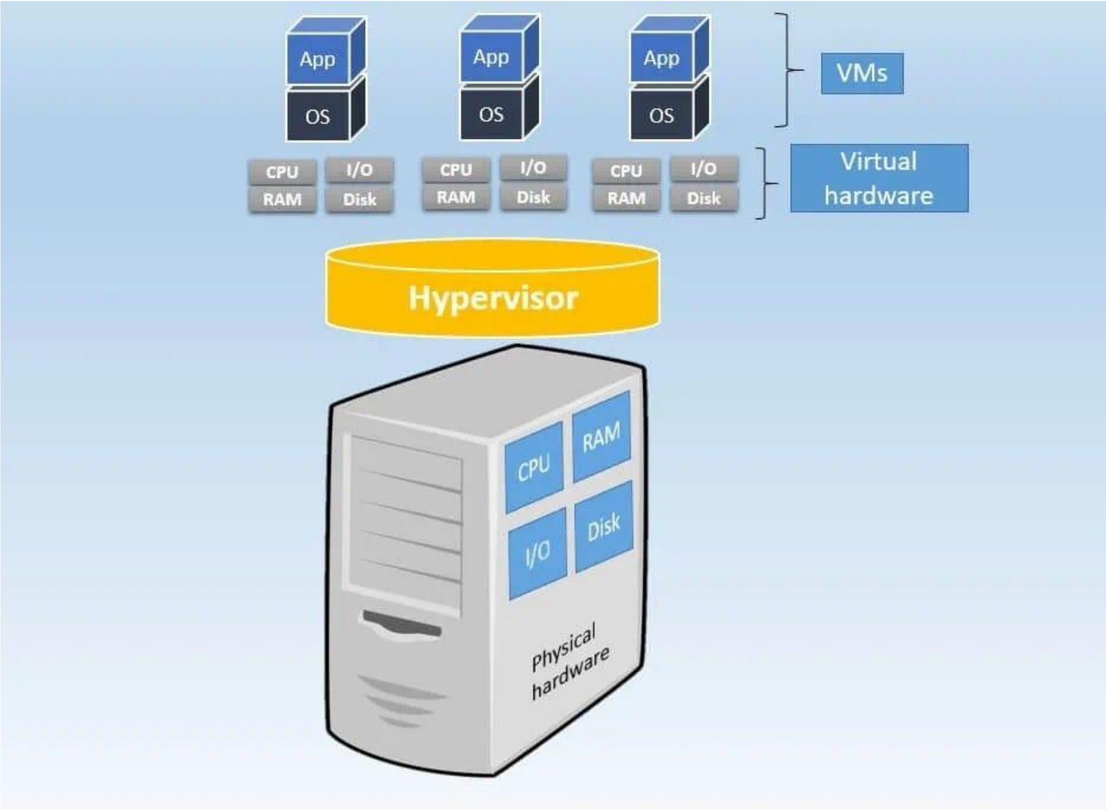
UNIT 2

Virtualization Architecture

The physical server




```
PHYSICAL SERVER
```

CPU RAM SSD / HDD Network Card

```
Virtual server example:
```

1. Student Information System
2. Library Management System
3. Examination System
4. Research Application

Virtualization

```
              ONE PHYSICAL SERVER
                       │
                       ▼
                 VIRTUALIZATION
                       │
       ┌───────────────┼───────────────┐
       ▼               ▼               ▼
      VM1             VM2             VM3        VM4
    Student         Library          Exam      Research
     System          System         System     System
```

"A Virtual Machine is like a computer created using software."

Hypervisor

```
     VM 1          VM 2          VM 3          VM 4
    Student       Library        Exam        Research
      App           App           App           App
       │             │             │             │
       ▼             ▼             ▼             ▼
   Guest OS      Guest OS      Guest OS      Guest OS
       └─────────────┬─────────────┴─────────────┘
                     ▼
                 HYPERVISOR
                     ▼
              PHYSICAL SERVER
```

The Architecture Through Four Resource Types

"The hypervisor virtualizes four major things: CPU, Memory, Storage, and Network."

```
                 HYPERVISOR
                      │
        ┌─────────────┼─────────────┐
        ▼             ▼             ▼
       CPU          MEMORY        STORAGE
   Virtual CPU    Virtual RAM   Virtual Disk
        │             │             │
        └─────────────┼─────────────┘
                      ▼
                   NETWORK
                 Virtual NIC
                      │
                      ▼
              Physical Hardware
```

A. CPU Virtualization

Imagine the physical server has:

16 CPU cores

We have four VMs.

```
Physical CPU = 16 cores

       Hypervisor
           │
   ┌───────┼─────────┐
   ▼       ▼         ▼
  VM1     VM2       VM3       VM4
 4vCPU   4vCPU     4vCPU     4vCPU
```

```
VM1 ─────┐
VM2 ─────┤
VM3 ─────┼────→ Hypervisor → Physical CPU
VM4 ─────┘
```

The hypervisor schedules CPU time.

example

- VM1 – Student Portal → 20% CPU
- VM2 – Library → 10% CPU
- VM3 – Exam System → 60% CPU
- VM4 – Research → 10% CPU

Memory Virtualization

Physical RAM = **128 GB**

```
VM1 – Student Portal → 32 GB
VM2 – Library → 16 GB
VM3 – Exam System → 48 GB
VM4 – Research → 32 GB
```

Total:

32 + 16 + 48 + 32 = 128 GB

The hypervisor manages the allocation.

```
             Physical RAM
                128 GB
```

```
         Hypervisor

VM1           VM2            VM3         VM4
32 GB         16 GB          48 GB       32 GB
```

"Virtualization is not only about sharing resources. It is also about controlled sharing and isolation."

Storage Virtualization

Suppose the physical server has:

2 TB SSD

The hypervisor provides virtual disks

```
Physical Storage
   2 TB SSD

     Hypervisor

  VM1    VM2    VM3    VM4
500GB  300GB  700GB  500GB
```

Network Virtualization

Now suppose all four VMs need internet access.

Each VM gets a **virtual network interface card (vNIC)**.

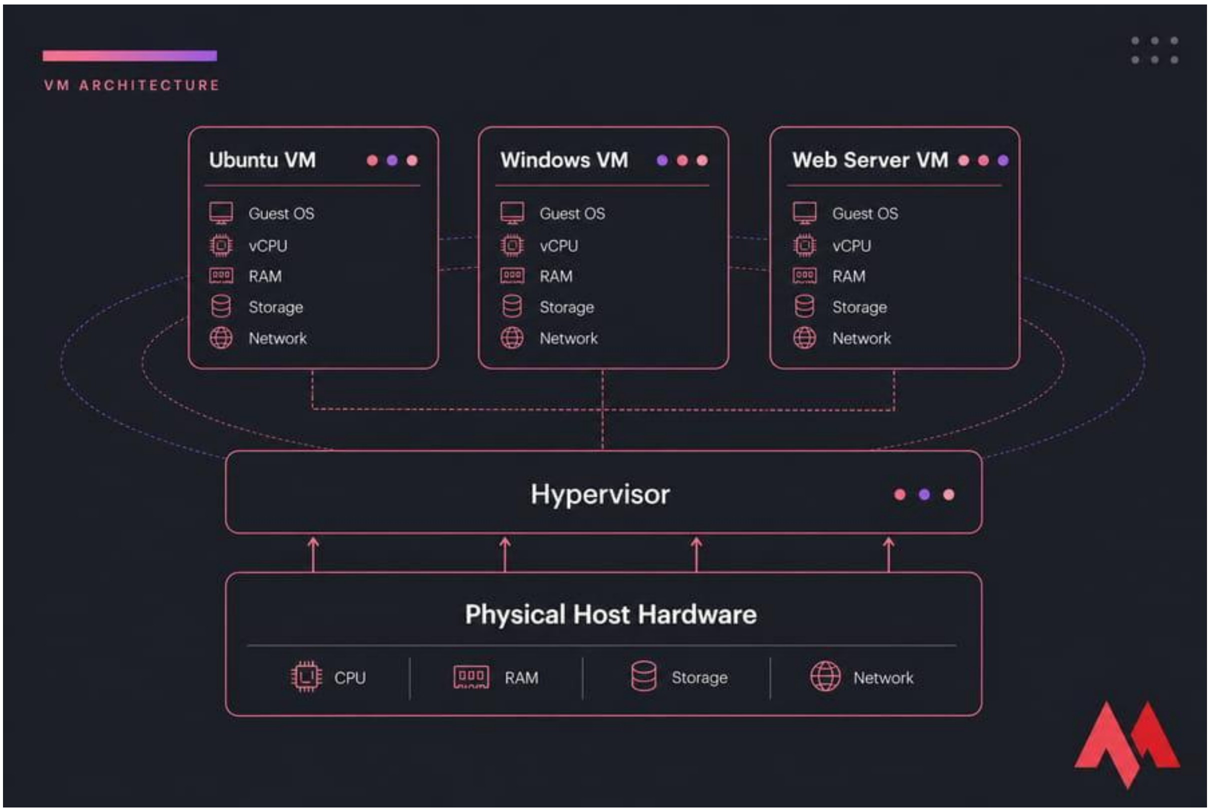
```
              Physical Network

             Physical NIC

             Virtual Switch

       VM1         VM2          VM3        VM4
      vNIC        vNIC         vNIC       vNIC
```

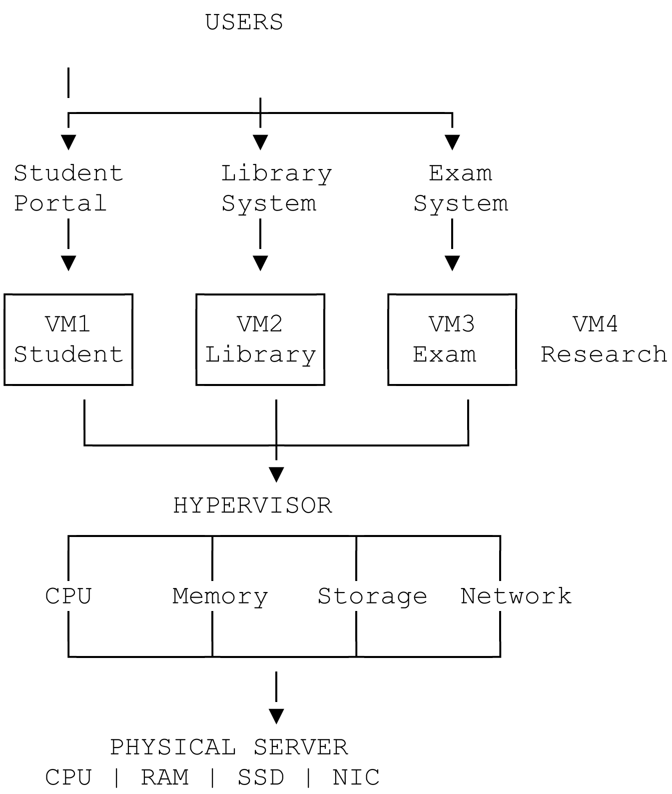
The **virtual switch** connects the VMs to each other and, depending on configuration, to the physical network.

example

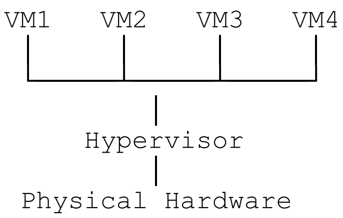
```
Student Portal VM
                 |
Library VM —— Virtual Switch —— Internet
                 |
Exam VM
                 |
Research VM
```

Complete architecture.





Type 1 vs Type 2



Type 1 — Bare Metal

"The hypervisor runs directly on the physical hardware."

Examples:

- VMware ESXi
- Microsoft Hyper-V
- Xen

Type 1 → Hypervisor first

```
VMs
 ↓
Hypervisor
 ↓
Hardware
```

Type 2:

```
Ubuntu VM
     |
VirtualBox
     |
Windows 11
     |
Laptop Hardware
```

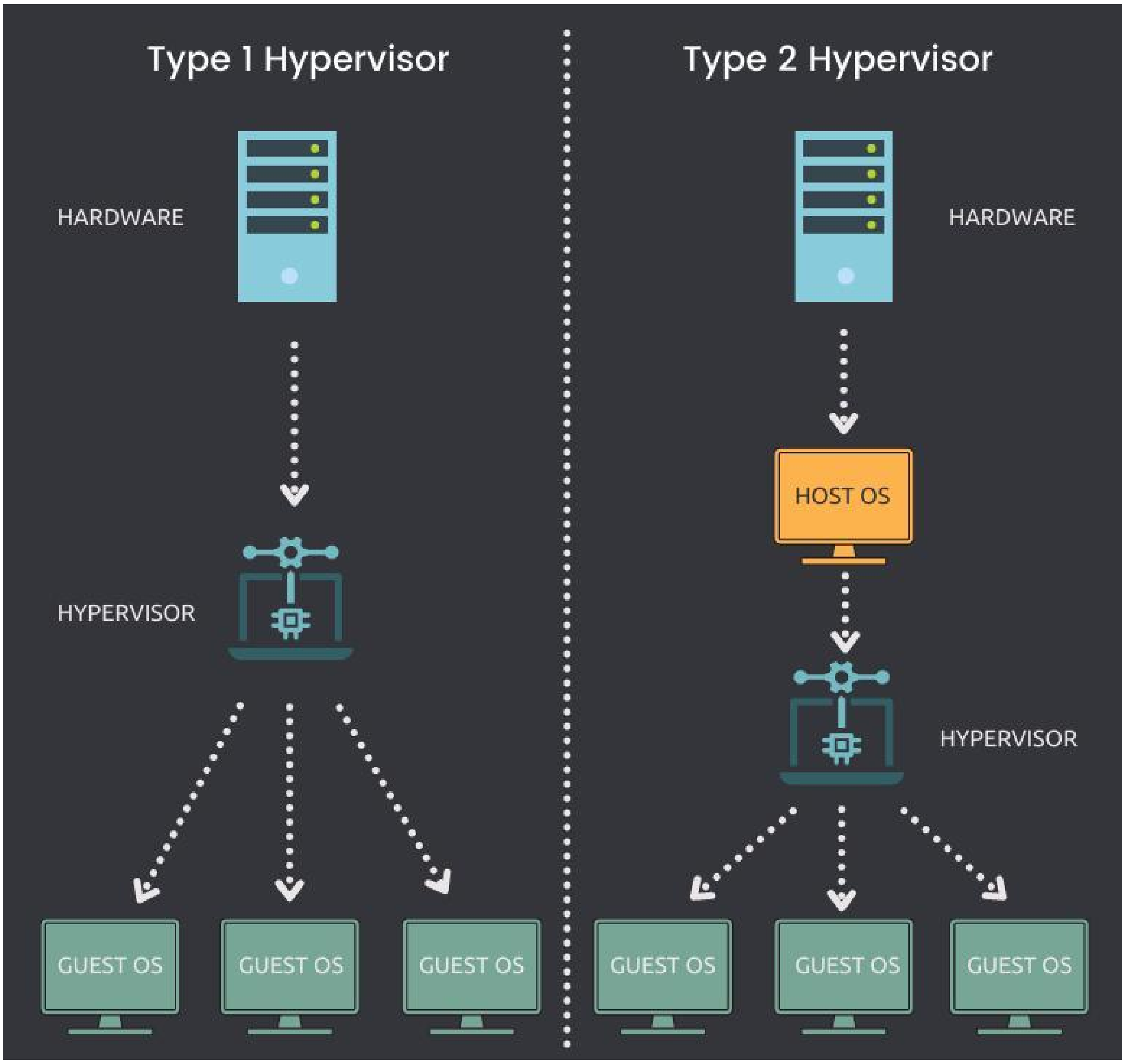
“the hypervisor is running on top of a host operating system."

Examples:

- VMware Workstation
- Oracle VirtualBox

Type 2 → Host OS first

```
VMs
 ↓
Hypervisor
 ↓
Host OS
 ↓
Hardware
```



Example scenartio

Case A

A student wants to run Ubuntu on a Windows laptop.

Type 1 or Type 2?

→ **Type 2**

Case B

A cloud provider wants to run thousands of customer VMs on physical servers.

Type 1 or Type 2?

→ **Type 1**

Case C

A developer wants to test a new application on Windows, Ubuntu, and Fedora using one laptop.

Type 1 or Type 2?

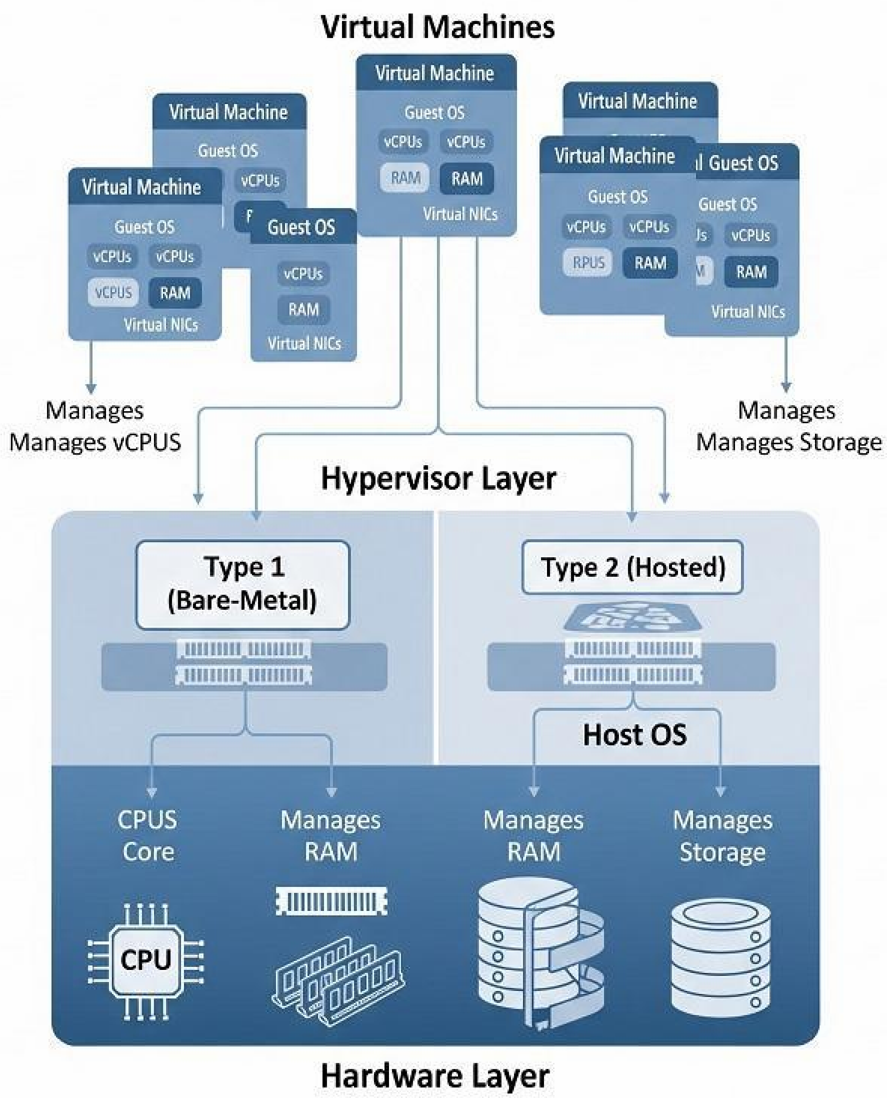
→ **Type 2**

Case D

A data center wants a dedicated virtualization layer directly on its servers.

Type 1 or Type 2?

→ **Type 1**



OS installation

Without virtualization

```
Physical Server
      └── One OS
            └── Applications
```

With virtualization

```
Physical Server
        │
   Hypervisor
        │
  ┌─────┼─────┐
VM1   VM2   VM3   VM4
 │     │     │     │
OS    OS    OS    OS
```

The key difference is that virtualization provides:

- **Isolation**
- **Resource sharing**
- **Resource management**
- **Independent operating environments**
- **Better server utilization**

SUMMARY

```
1. ONE PHYSICAL SERVER
          ↓
2. TOO MANY WORKLOADS
          ↓
3. HOW CAN ONE SERVER RUN MANY SYSTEMS?
          ↓
4. VIRTUAL MACHINE
          ↓
5. WHO MANAGES THE VMs?
          ↓
6. HYPERVISOR
          ↓
7. WHAT DOES IT MANAGE?
          ↓
   CPU | MEMORY | STORAGE | NETWORK
          ↓
8. WHERE DOES HYPERVISOR RUN?
          ↓
   TYPE 1 vs TYPE 2
```

MCQs: Virtualization Architecture

Q1. [Architecture]

A physical server has 16 CPU cores and 64 GB RAM. A hypervisor hosts four VMs, each configured with 4 vCPUs and 16 GB RAM. Which statement is correct?

A. The hypervisor must permanently dedicate exactly 4 physical cores and 16 GB RAM to each VM.
B. Each VM can use virtual CPU and memory resources, while the hypervisor manages their mapping to physical resources.
C. All VMs must execute on separate physical CPUs simultaneously.
D. Virtualization eliminates the need for physical CPU and memory.

Answer: B

Explanation:

A VM sees **virtual hardware**. The hypervisor maps virtual CPUs to physical CPU resources and manages memory allocation. The mapping need not always be a fixed one-to-one assignment.

Q2. [Type 1 vs Type 2]

Consider the following architectures:

Architecture I

```
VMs
 ↓
Hypervisor
 ↓
Hardware
```

Architecture II

```
VMs
 ↓
Hypervisor
 ↓
Host OS
 ↓
Hardware
```

Which statement is correct?

A. Architecture I represents Type 2 virtualization.
B. Architecture II represents Type 1 virtualization.
C. Architecture I represents Type 1 and Architecture II represents Type 2.
D. Both represent Type 1 virtualization.

Answer: C

Explanation:

A **Type 1 hypervisor** runs directly on hardware. A **Type 2 hypervisor** runs above a host operating system.

Q3. [Resource Allocation]

A hypervisor runs three VMs. VM1 requires 80% CPU utilization, VM2 requires 10%, and VM3 requires 10%. If the physical CPU becomes heavily loaded, the hypervisor's primary responsibility is to:

A. Give all CPU time permanently to VM1.
B. Schedule CPU resources among VMs according to its resource management policies.
C. Shut down VM2 and VM3 automatically in all cases.
D. Create additional physical CPUs.

Answer: B

Explanation:

The hypervisor performs **resource scheduling**. CPU time can be dynamically allocated based on configured policies, priorities, reservations, limits, or shares.

Q4. [Isolation]

Three independent VMs run on the same physical host. VM1 crashes because of an operating-system error. Assuming correct hypervisor isolation, what is the most appropriate outcome?

A. The physical server must always crash.
B. All other VMs must necessarily crash.
C. VM1 may become unavailable while other VMs continue running.
D. The hypervisor must automatically reinstall VM1.

Answer: C

Explanation:

One of the important properties of virtualization is **isolation**. A failure inside one VM should generally not directly corrupt the execution environment of other VMs.

Q5. [Virtual CPU]

A VM is configured with 4 vCPUs on a physical host containing 2 physical CPU cores. Which statement is most accurate?

A. The VM can never execute because vCPUs must equal physical cores.
B. The hypervisor can schedule the 4 vCPUs over the available physical CPU resources.
C. Four physical CPU cores are automatically created.
D. The guest OS directly controls the physical CPU allocation.

Answer: B

Explanation:

Virtualization allows a VM to see a configured number of **vCPUs**. The hypervisor schedules these virtual CPUs onto available physical CPU resources. However, allocating more vCPUs than physical cores can introduce scheduling overhead and contention.

Q6. [Memory Virtualization]

A physical host has 16 GB RAM. Two VMs are configured with 8 GB RAM each. Which statement is correct?

A. The VMs cannot share the physical host.
B. The hypervisor manages the mapping of guest virtual memory to physical memory.
C. Each VM receives a separate physical RAM chip.
D. The guest operating system directly allocates physical RAM without hypervisor involvement.

Answer: B

Explanation:

The guest OS operates with the abstraction of memory provided to the VM. The hypervisor manages the mapping between **guest/virtual memory** and **physical machine memory**.

Q7. [Virtual Network]

Which component is commonly used to connect multiple VMs to a virtualized network environment?

A. Virtual switch
B. Compiler
C. BIOS
D. File system

Answer: A

Explanation:

A **virtual switch** provides virtual network connectivity between VMs and, depending on configuration, the physical network. Each VM may have a virtual network interface card (vNIC).

Q8. [Virtualization Architecture]

Consider:

```
Applications
     ↓
Guest OS
     ↓
Virtual Hardware
     ↓
Hypervisor
     ↓
Physical Hardware
```

Which layer primarily provides the abstraction of CPU, memory, storage, and network resources to the guest OS?

A. Physical hardware
B. Hypervisor
C. Application layer
D. User interface

Answer: B

Explanation:

The **hypervisor** presents virtualized hardware resources to guest operating systems and manages their access to underlying physical resources.

Q9. [

Conceptual]

Which of the following is **NOT** a fundamental goal or benefit of server virtualization?

A. Server consolidation
B. Resource isolation
C. Improved hardware utilization
D. Guaranteed elimination of all performance overhead

Answer: D

Explanation:

Virtualization can improve utilization and provide isolation and consolidation. However, it does **not guarantee zero performance overhead**. Virtualization itself may introduce overhead due to resource management and virtualization operations.

Q10. [Architecture Analysis]

A company wants to run multiple isolated operating systems directly on a physical data-center server without installing a conventional host OS beneath the virtualization layer. Which architecture is most appropriate?

A. Type 1 hypervisor
B. Type 2 hypervisor
C. Dual-boot system
**D. Container-only architecture

Answer: A

Explanation:

A **Type 1 (bare-metal) hypervisor** runs directly on the physical hardware and manages multiple guest VMs. This architecture is widely associated with enterprise server and data-center virtualization.

Security problems in Virtualization

```
          PHYSICAL SERVER
     CPU | RAM | SSD | Network
                |
                ▼
                HYPERVISOR
          ┌─────────┼─────────┐
          ▼         ▼         ▼
     VM 1      VM 2      VM 3
     Student   Exam    Library
     System    System    System
```

What is Isolation?

Normally, each VM should be separated from the others.

In virtualization:

```
VM1  ✕  VM2  ✕  VM3
```

The **hypervisor provides isolation**.

So the first security goal is:

VM1 should not be able to directly access VM2's data or memory.

#1: VM Escape

```
VM1
 |
 | Cannot access directly
 ▼
Hypervisor
 |
 ├── VM1
 ├── VM2
 └── VM3
```

But imagine there is a **security vulnerability in the virtualization software**.

An attacker inside VM1 finds a way to break out of the VM.

This is called:

VM Escape

```
Normal:

VM1 → Hypervisor → Physical Hardware
```

```
Attack:

VM1 → Escape vulnerability
          ↓
     Hypervisor
          ↓
   Other VMs / Host
```

VM escape is a security attack in which an attacker breaks out of a virtual machine and potentially gains access to the host or other virtualized resources.

#2: Hypervisor Attack

The hypervisor is the **most important middle layer**.

```
          VM1
          |
          VM2
          |
          VM3
          |
          ▼
     HYPERVISOR
          |
          ▼
     PHYSICAL HARDWARE
```

Every VM depends on the hypervisor.

So imagine an attacker attacks the hypervisor itself.

```
          Attacker
          |
          ▼
     HYPERVISOR
     ┌────┼────┐
     ▼    ▼    ▼
     VM1  VM2  VM3
```

If the hypervisor is compromised, the attacker could potentially affect multiple VMs.

That's why the hypervisor must be:

- Updated
- Patched
- Properly configured
- Protected with strong access controls

#3: VM-to-VM Attack

Suppose we have:

```
VM1 = Attacker
VM2 = Student Database
```

Both are running on the same physical server.

The attacker tries to attack VM2.

```
┌───────────┐          ┌─────────────┐
│    VM1    │ ───────► │   VM2    │
│ Attacker  │  Attack  │ Database       │
└───────────┘          └─────────────┘
```

This is a **VM-to-VM attack**.

The attacker may try:

- Network attacks
- Exploiting vulnerabilities
- Stealing credentials
- Exploiting weak configurations

Virtual machines are isolated, but they may still communicate over **virtual networks**.

This is why we need:

- Firewalls
- Network segmentation
- Access control
- Authentication
- Encryption
- Intrusion detection

#4: Insecure VM Image

```
          VM IMAGE

             │
    ┌───────┼───────┐
    ▼       ▼         ▼
   VM1     VM2       VM3
```

Imagine the original VM image contains malware.

Then:

```
Malware Image
        |
        ├── VM1 → Infected
        ├── VM2 → Infected
        └── VM3 → Infected
```

This is why VM images should be:

- Trusted
- Scanned
- Updated
- Patched

#5: VM Sprawl

Eg

```
VM1
VM2
VM3
VM4
...
VM100
```

Some VMs are no longer being used.

But nobody deletes them.

```
Old VM
    |
    ├── Outdated OS
    ├── Old software
    ├── Unpatched vulnerabilities
    └── Forgotten administrator password
```

An attacker may discover this forgotten VM.

This is called:

VM Sprawl

VM sprawl = uncontrolled growth of virtual machines that are difficult to manage and secure.

#6: Resource Exhaustion

Now imagine one VM consumes too many resources.

```
Physical Server
        |
   Hypervisor
```

```
          │
    ┌─────┼─────┐
    ▼     ▼    ▼
VM1   VM2    VM3
 │
 │
 ▼
Uses 95% CPU
```

VM1 is consuming almost all the CPU.

Then:

```
VM1 → 95% CPU
VM2 → Very slow
VM3 → Very slow
```

This can cause a **Denial of Service (DoS)-like effect** on other VMs.

The hypervisor can help control this using:

- CPU limits
- Memory limits
- Resource quotas
- Resource reservations

Overall architecture

```
                  PHYSICAL SERVER
                        │
                  HYPERVISOR
                        │
      ┌───────────────┼───────────────┐
      ▼                 ▼                 ▼
     VM1               VM2               VM3
   Attacker         Database         Library
      │
      │
      ├── VM Escape
      │
      ├── VM-to-VM Attack ────────▶ VM2
      │
      └── Resource Exhaustion
```

```
Hypervisor itself
        |
        └── Hypervisor Attack

VM Images
        |
        └── Malware / Vulnerabilities

Too many VMs
        |
        └── VM Sprawl
```

Example

"A cloud provider has one physical server hosting three customers."

```
Physical Server
       |
  Hypervisor
       |
 ┌─────┼─────┐
 ▼     ▼     ▼
VM1   VM2   VM3
A      B     C
```

Now tell students:

Question 1

Customer A's VM is hacked. What if the attacker tries to leave VM1?

→ VM Escape

Question 2

The attacker tries to attack Customer B's VM.

→ VM-to-VM Attack

Question 3

The attacker targets the virtualization software itself.

→ Hypervisor Attack

Question 4

The administrator creates 1,000 VMs and forgets to remove old ones.

→ VM Sprawl

Question 5

One VM consumes all CPU and RAM.

→ **Resource Exhaustion**

Question 6

The administrator uses a VM image containing malware.

→ **Insecure VM Image**

Cloud Data Center Architecture

1. What is a Data Center?

A **data center** is a physical facility that contains a large number of:

- Servers
- Storage systems
- Network devices
- Security devices
- Power systems
- Cooling systems

For example, a cloud provider operates huge data centers containing thousands of servers.

Eg

college wants to create a **Cloud-Based Student Management System**.

It needs:

- A server to run the application
- Storage to store student data
- Network connectivity for students and staff
- Security to protect the system
- Virtualization to efficiently use physical servers

The basic architecture is:

```
  STUDENTS / FACULTY
          │
          │ Internet
          ▼
 ┌────────────────┐
 │   CLOUD        │
 │  DATA CENTER   │
```

```
         └────────────────┘
                 │
  ┌──────────────┼───────────────┐
  ▼           ▼            ▼
Compute                 Storage     Network
  │           │            │
  ▼           ▼            ▼
Virtual                 Data        Virtual
Machines                Storage     Networks
```

2. Basic Cloud Data Center Architecture

A simplified architecture can be shown as:

```
                    USERS
                      │
                      ▼
                    INTERNET
                      │
                      ▼
              EDGE / ROUTER
                      │
                      ▼
              LOAD BALANCER
                      │
                      ▼
        ┌────────────────────────┐
           │   CLOUD DATA CENTER    │
        │                        │
        │  ┌──────────────────┐  │
        │  │ Compute Layer    │  │
        │  │ Servers + VMs    │  │
        │  └──────────────────┘  │
        │          │           │
        │  ┌───────┴──────────┐  │
        │  │               │  │
        │  ▼               ▼  │
        │ Storage         Network
        │                 │  │
        └────────────────────────┘
```

Layer 1 – Physical Infrastructure

```
  PHYSICAL INFRASTRUCTURE
              │
  ┌───────────┼─────────────┐
  ▼           ▼             ▼
Servers     Storage       Network
  │           │             │
  ▼           ▼             ▼
CPU/RAM     Disks         Switches
```

```
                    Routers
                    Firewalls
```

Compute

Physical servers contain:

- CPU
- RAM
- Motherboard
- Network interfaces

Storage

Used to store:

- VM disks
- Databases
- Files
- Images
- Backups

Network

Connects:

- Servers
- Storage
- VMs
- Users
- Internet

Layer 2 – Virtualization Layer

Physical servers run **hypervisors**.

```
        PHYSICAL SERVER
              │
              ▼
        HYPERVISOR
              │
        ┌─────┼──────────┐
        ▼          ▼          ▼
    VM1        VM2       VM3
    Windows    Linux     Linux
```

The hypervisor converts physical resources into virtual resources:

Physical Resource	Virtual Resource
Physical CPU	vCPU
Physical RAM	Virtual Memory

Physical Disk	Virtual Disk
Physical NIC	Virtual NIC

Layer 3 – Compute

The **compute layer** provides processing power.

For example:

```
Physical Server 1
        │
   Hypervisor
        │
   ┌────┼────┐
   ▼    ▼    ▼
  VM1  VM2  VM3

Physical Server 2
        │
   Hypervisor
        │
   ┌────┼────┐
   ▼    ▼    ▼
  VM4  VM5  VM6
```

The cloud management system can decide where a VM should run.

For example:

"Server 1 is overloaded. Create the new VM on Server 2."

This is called **resource management**.

Layer 4 – Storage

A cloud data center needs massive storage.

Instead of every server having only its own local disk, cloud environments often use shared or distributed storage systems.

Conceptually:

```
        VM1          VM2        VM3
        │            │          │
        └──────────┼─────────┘
                     │
                     ▼
            Cloud Storage System
                     │
        ┌──────────┼──────────┐
        ▼            ▼          ▼
```

```
      Disk 1     Disk 2       Disk 3
```

Cloud storage may provide:

Block Storage

Used like a virtual hard disk.

```
VM → Virtual Disk
```

File Storage

Used for shared files.

```
VM1 ──┐
VM2 ──┼──► Shared File System
VM3 ──┘
```

Object Storage

Used for large amounts of unstructured data such as:

- Images
- Videos
- Documents
- Backups

Layer 5 – Network Infrastructure

Networking connects everything.

A simplified flow is:

```
User
 |
Internet
 |
Router
 |
Firewall
 |
Load Balancer
 |
Virtual Network
 |
Virtual Machine
```

Inside the data center:

```
VM1 ──┐
VM2 ──┼── Virtual Switch
VM3 ──┘
            |
            ▼
      Physical Switch
            |
```

```
           ▼
      Router
```

This is where the concept from your previous question fits.

VMs can be isolated at the **resource level**, but they can communicate through **virtual networks** when permitted.

Layer 6 – Security

Security is present throughout the architecture.

```
                  CLOUD DATA CENTER
                        │
      ┌─────────────────┼─────────────────┐
      ▼                 ▼                 ▼
      Compute           Storage           Network
      │                 │                 │
      VM Security       Data Security     Network Security
      │                 │                 │
      └─────────────────┼─────────────────┘
                        ▼
                        Security
```

Security mechanisms can include:

- Firewalls
- Identity and access management
- Encryption
- Network segmentation
- Intrusion detection
- VM isolation
- Secure VM images

For example:

```
VM1 ──X──► VM2
      Blocked

VM1 ──────► Database
      Allowed only on required port
```

Layer 7 – Cloud Management and Orchestration

This is the **brain of the cloud data center**.

It manages:

- VM creation

- VM deletion
- Resource allocation
- Monitoring
- Scaling
- User management
- Security policies
- Automation

Suppose a user requests:

"Create a VM with 4 vCPUs, 16 GB RAM, and 100 GB storage."

The management layer coordinates:

```
User Request
      |
      ▼
Cloud Management
      |
      ├── Select Compute
      |
      ├── Allocate CPU/RAM
      |
      ├── Allocate Storage
      |
      ├── Configure Network
      |
      └── Apply Security
                |
                ▼
            New VM
```

This is a key difference between **simple virtualization** and **cloud computing**.

A hypervisor manages VMs on a host.

A cloud management platform coordinates **many physical servers, virtual machines, networks, and storage resources**.

Complete Cloud Data Center Architecture

Now combine everything:

```
            CLOUD USERS
      /     |       \
      /     |        \
      ▼           ▼           ▼
Web Browser  Mobile App  API
      \     |        /
      \     |       /
            INTERNET
                  |
                  ▼
      ROUTER / EDGE
                  |
                  ▼
            FIREWALL
                  |
                ▼
      LOAD BALANCER
                  |
                  ▼
┌────────────────────────────┐
|     CLOUD DATA CENTER        |
|                              |
|  ┌─────────────────────┐    |
|  | Cloud Management  |   |
|  | & Orchestration   |   |
|  └──────────┬──────────┘    |
|           |           |
|  ┌──────────┼──────────┐    |
|  ▼        ▼           ▼   |
|Compute    Storage    Network
|  |        |           |  |
|  ▼        ▼           ▼  |
|Hypervisor  Storage   Virtual
|  |        System      Network
|  |                    |  |
|  ▼                    ▼  |
|Virtual                 Physical
|Machines                Network
|                              |
└──────────────┬─────────────┘
                  |
                  ▼
      Physical Hardware
```

SDDC

Eg

college has a data center:

```
Physical Servers
Physical Storage
Physical Network
```

In a traditional data center, many operations require manual configuration of physical infrastructure.

This can be:

- Slow
- Complex
- Difficult to scale
- Expensive to manage

So the industry moved toward **software-defined infrastructure**.

2. What is an SDDC?

Software-Defined Data Center (SDDC)

An **SDDC is a data center in which computing, storage, networking, and security infrastructure are virtualized and managed through software.**

The main idea is:

"Instead of manually managing every physical hardware component, we use software to create, configure, control, and automate infrastructure resources."

A simple definition for students:

SDDC = A data center where infrastructure is abstracted from physical hardware and controlled through software.

3. Traditional Data Center vs SDDC

Traditional Data Center

```
            DATA CENTER
                 │
     ┌───────────┼───────────┐
     ▼           ▼           ▼
   Servers     Storage     Network
     │           │           │
     ▼           ▼           ▼
   Physical    Physical   Physical
   Hardware    Hardware   Hardware
```

Suppose you want to:

"Create a new server."

The administrator may need to:

1. Purchase a physical server
2. Install it
3. Connect network cables
4. Install an operating system
5. Configure storage
6. Configure networking

This takes time.

SDDC

In an SDDC:

```
            SOFTWARE
                 │
     ┌───────────┼───────────┐
     ▼           ▼           ▼
  Compute     Storage    Network
     │           │           │
     ▼           ▼           ▼
Virtualized  Virtualized Virtualized
     │           │           │
     └───────────┼───────────┘
                 ▼
      Physical Infrastructure
```

An administrator can use software to provision and manage resources.

For example:

"Create a new VM with 4 vCPUs, 16 GB RAM, and 100 GB storage."

The system can automatically:

- Select a physical server

- Create the VM
- Allocate CPU and memory
- Allocate virtual storage
- Configure virtual networking
- Apply security policies

This is the power of **software-defined infrastructure**.

4. The Three Main Components

```
           SDDC
              │
     ┌──────────┼──────────┐
     ▼          ▼           ▼
  Compute     Storage   Networking
 Virtual      Virtual   Virtual
```

5. Software-Defined Compute

Traditional:

```
Physical Server
      │
      ▼
Operating System
      │
      ▼
Application
```

Virtualized:

```
Physical Server
      │
      ▼
Hypervisor
      │
 ┌────┼────┐
 ▼    ▼     ▼
VM1  VM2  VM3
```

In SDDC, compute resources are managed through software.

The administrator can:

- Create VMs
- Delete VMs
- Resize VMs
- Move VMs
- Allocate CPU
- Allocate memory

For example:

```
VM1
4 vCPU
16 GB RAM
```

If the application needs more resources:

```
Before:
4 vCPU
16 GB RAM

After:
8 vCPU
32 GB RAM
```

The resources can be adjusted through software without physically replacing the server.

Software-defined compute = Physical computing resources are abstracted and provided as software-controlled virtual resources.

Software-Defined Storage

imagine a data center has:

```
Storage 1 → 10 TB
Storage 2 → 20 TB
Storage 3 → 30 TB
```

Total:

```
60 TB
```

Instead of managing each storage device separately, the storage can be presented as a **logical storage pool**.

```
Physical Storage
 ┌────────┬────────┬────────┐
 │ 10TB   │ 20TB   │ 30TB   │
 └────────┴────────┴────────┘
                │
                ▼
      Storage Virtualization
                │
                ▼
      Storage Pool
                │
      ┌─────────┼─────────┐
      ▼         ▼         ▼
      VM1      VM2      VM3
```

The software manages the storage resources.

For example:

VM1 needs 100 GB.

The system can allocate 100 GB from the available storage pool.

The administrator does not necessarily need to manually identify a specific physical disk.

Software-defined storage = Storage resources are abstracted into logical pools and managed through software.

Software-Defined Networking

Traditional networking:

```
Computer
   |
Cable
   |
Physical Switch
   |
Router
```

Configuration is often tied closely to physical networking devices.

In SDDC:

```
Application
      |
      ▼
Virtual Network
      |
      ▼
Virtual Switch
      |
      ▼
Virtual Router
      |
      ▼
Physical Network
```

The network can be configured through software.

For example:

```
VM1 ─────────▶ VM2
  Allowed

VM1 ──X───────▶ Database
  Blocked
```

The administrator can define network policies through software.

This includes:

- Virtual switches
- Virtual routers
- Virtual firewalls
- Network segmentation
- Access policies

Software-defined networking = Network resources and policies are controlled programmatically through software rather than relying solely on manual configuration of physical devices.

The SDDC Architecture

combine all three.

```
                  SDDC
                   │
            Software Management Layer
                   │
     ┌─────────────┼─────────────┐
     ▼             ▼             ▼
  Compute       Storage       Network
     │             │             │
     ▼             ▼             ▼
 Hypervisor     Storage       Virtual
     │       Virtualization   Network
     │             │             │
     └─────────────┼─────────────┘
                   │
                   ▼
            Physical Infrastructure
     ┌─────────────┼─────────────┐
     ▼             ▼             ▼
  Servers       Storage       Network
```

The physical hardware still exists. SDDC does not eliminate hardware. It abstracts the hardware and allows software to manage it.

What Does "Software-Defined" Actually Mean?

```
Traditional:

Administrator
      ↓
Physical Hardware
      ↓
Manual Configuration


SDDC:

Administrator
      ↓
Software Management
```

```
      ↓
Virtual Resources
      ↓
Physical Hardware
```

Note:

Software-defined does not mean hardware-free.

Hardware resources are abstracted and controlled through software.

How SDDC Relates to Cloud Computing

```
Physical Infrastructure
      │
      ▼
Virtualization
      │
      ▼
Hypervisors
      │
      ▼
SDDC
 ┌──────┼─────┐
 ▼      ▼   ▼
Compute Storage Network
      │
      ▼
Automation & Management
      │
      ▼
Cloud Data Center
      │
      ▼
Cloud Computing Services
```

Virtualization

Creates virtual versions of physical resources.

SDDC

Virtualizes and manages compute, storage, networking, and security through software.

Cloud Computing

Delivers those resources as services to users over a network.

BL3 – Apply (Q1–Q7)

Q1. [BL3]

A cloud data center has multiple physical servers running several virtual machines. A new VM needs 4 vCPUs, 16 GB RAM, and 200 GB storage. Which combination of components is primarily responsible for providing these resources?

A. Firewall, router, and load balancer
B. Hypervisor, compute infrastructure, and storage system
C. DNS server, firewall, and router
D. Virtual switch, firewall, and authentication server

Answer: B

Explanation: The **hypervisor** creates and manages VMs, the **compute infrastructure** provides CPU and memory, and the **storage system** provides virtual disk capacity.

Q2. [BL3]

A cloud provider wants to allow VM1 and VM2 to communicate with each other while preventing VM1 from directly accessing VM2's memory and virtual disk. Which approach best achieves this?

A. Remove virtualization from the host
B. Allow direct hardware access between VMs
C. Use hypervisor-based resource isolation and controlled virtual networking
D. Store both VMs on the same physical disk without isolation

Answer: C

Explanation: The hypervisor isolates CPU, memory, and storage resources, while a **virtual network** can provide controlled communication between VMs.

Q3. [BL3]

An organization wants to create a new application server within minutes without purchasing or physically installing a new server. Which SDDC capability enables this?

A. Manual hardware provisioning
B. Software-defined compute and virtualization
C. Physical network cabling
D. Dedicated storage hardware for every application

Answer: B

Explanation: Software-defined compute uses virtualization to create VMs from existing physical resources, allowing rapid provisioning through software.

Q4. [BL3]

A cloud administrator wants to provide 500 GB of virtual storage to a VM without specifying a particular physical disk. Which SDDC technology is most appropriate?

A. Software-defined storage
B. Physical server clustering
C. Hardware firewall
D. Load balancing

Answer: A

Explanation: Software-defined storage (SDS) abstracts physical storage into logical storage pools and allocates virtual storage resources as required.

Q5. [BL3]

A cloud provider wants to create isolated virtual networks for finance, healthcare, and student applications running on the same physical infrastructure. Which SDDC capability is most appropriate?

A. Software-defined networking
B. Physical CPU virtualization
C. Manual disk partitioning
D. VM cloning only

Answer: A

Explanation: Software-defined networking (SDN) enables logical networks, segmentation, virtual switches, routers, and security policies to be configured through software.

Q6. [BL3]

A malicious user has compromised VM1 and attempts to access the memory of VM2 directly. Which security mechanism should prevent this?

A. Load balancing
B. Hypervisor-enforced memory isolation
C. Storage virtualization
D. DNS resolution

Answer: B

Explanation: The hypervisor maintains isolation between VM memory spaces. A VM should not directly access another VM's memory.

Q7. [BL3]

A cloud administrator discovers that an old VM has not been patched for two years and is no longer being used. Which virtualization security problem does this best represent?

A. VM escape
B. VM sprawl
C. Network virtualization
D. Storage pooling

Answer: B

Explanation: VM sprawl occurs when organizations create and retain too many VMs that become difficult to track, manage, patch, and secure.

BL4 – Analyze (Q8–Q14)

Q8. [BL4]

Consider the following architecture:

```
Users
  ↓
Internet
  ↓
Load Balancer
  ↓
VM1    VM2    VM3
  ↓      ↓      ↓
       Database
```

VM1, VM2, and VM3 are running on different physical hosts but access the same database service. Which statement best explains this architecture?

A. VMs must always share the same physical server to communicate
B. VMs can be physically isolated while communicating through logical network connections
C. VMs cannot communicate if they are isolated
D. Database access requires direct memory sharing between VMs

Answer: B

Explanation: Resource isolation does not prevent network communication. VMs can run on separate hosts and communicate through physical or virtual networks using controlled network connections.

Q9. [BL4]

A traditional data center requires an administrator to manually configure servers, storage devices, and physical network switches. An SDDC automates these operations through software. Which architectural change primarily explains the improvement?

A. Elimination of all physical hardware
B. Abstraction and software-based control of infrastructure resources

C. Replacement of all servers with routers
D. Removal of virtualization

Answer: B

Explanation: SDDC does not eliminate hardware. Instead, it **abstracts compute, storage, networking, and security resources** and enables them to be managed through software.

Q10. [BL4]

A cloud data center experiences the following conditions:

```
CPU utilization = 93%
Memory utilization = 91%
Storage latency = 40 ms
VM startup time = 180 seconds
```

Which is the most likely explanation?

A. The network is completely idle
B. The host is experiencing multi-resource contention
C. The VMs have excessive CPU capacity
D. Storage latency cannot affect VM startup time

Answer: B

Explanation: High CPU, memory, and storage latency indicate **resource contention**. Storage latency can directly affect VM startup because VM images and virtual disks must be read during boot.

Q11. [BL4]

A company runs finance, healthcare, and student VMs on the same physical server. An attacker compromises the student VM and attempts to attack the finance VM through network communication. Which security mechanism should primarily reduce this risk?

A. Network segmentation and firewall policies
B. Increasing CPU frequency
C. Increasing VM disk size
D. VM cloning

Answer: A

Explanation: Network segmentation and firewall rules can restrict communication between security domains. Resource isolation alone does not prevent network-based attacks.

Q12. [BL4]

An attacker compromises a VM and then exploits a vulnerability in a virtualization component to gain access beyond the VM boundary. Which sequence best describes the attack?

A. VM compromise → VM escape → potential host compromise
B. Hypervisor installation → VM creation → VM escape
C. Storage pooling → network segmentation → VM escape
D. Load balancing → VM migration → VM escape

Answer: A

Explanation: The attacker first gains control of a VM and then exploits a vulnerability in the virtualization boundary, potentially escaping the VM and affecting the host or protected resources.

Q13. [BL4]

An SDDC administrator wants to move a workload from an overloaded physical host to another host with available resources. Which combination of capabilities is most relevant?

A. Virtualization and centralized resource management
B. Physical cabling and manual server replacement
C. DNS and email services
D. Data encryption alone

Answer: A

Explanation: Virtualization allows workloads to be abstracted from physical hardware, while centralized management helps monitor and allocate resources across hosts.

Q14. [BL4]

A cloud provider has a large storage pool consisting of multiple physical storage devices. Users see only logical virtual disks rather than individual physical disks. Which architectural principle is being applied?

A. Network isolation
B. Storage abstraction
C. VM escape
D. Physical server isolation

Answer: B

Explanation: Storage abstraction hides the underlying physical storage and presents logical storage resources to users and VMs. This is a core concept of software-defined storage.

BL5 – Evaluate (Q15–Q20)

Q15. [BL5]

A cloud provider must host mission-critical applications requiring high performance, strong isolation, and centralized management. Which architecture is the most appropriate?

A. Type-2 hypervisor running on a desktop operating system
B. Type-1 hypervisor running directly on physical hardware
C. Application running without virtualization
D. Multiple applications running directly on the host OS

Answer: B

Explanation: A **Type-1 hypervisor** runs directly on hardware and is generally preferred for enterprise and cloud data centers due to its performance, isolation, and data-center management capabilities.

Q16. [BL5]

A company claims that implementing an SDDC eliminates the need for physical servers, storage devices, and network hardware. Which evaluation is correct?

A. Correct, because SDDC is completely hardware-free
B. Correct, because all data centers become virtual machines
C. Incorrect, because SDDC abstracts and controls physical infrastructure through software
D. Incorrect, because SDDC does not use virtualization

Answer: C

Explanation: SDDC still relies on physical infrastructure. The key difference is that **software abstracts and manages the physical resources**, making them more programmable and automated.

Q17. [BL5]

A cloud provider wants to improve agility compared with a traditional data center. It wants to provision VMs, allocate storage, configure networks, and apply security policies automatically. Which approach is best?

A. Increase manual configuration of physical devices
B. Adopt an SDDC architecture with software-defined compute, storage, networking, and security
C. Eliminate virtualization
D. Assign one physical server to every application

Answer: B

Explanation: SDDC enables centralized software-based control and automation across **compute, storage, network, and security**, improving provisioning speed and operational agility.

Q18. [BL5]

A VM hosting a sensitive financial application is compromised. The attacker attempts to access another VM's memory and then tries to exploit the hypervisor. Which security architecture provides the strongest defense?

A. No isolation because all VMs need maximum connectivity
B. Hypervisor isolation combined with regular patching, network segmentation, and least privilege
C. Only increasing VM storage capacity
D. Only using a load balancer

Answer: B

Explanation: A layered security approach is necessary. **Hypervisor isolation** protects VM resources, while **patching** reduces hypervisor vulnerabilities, **network segmentation** limits lateral movement, and **least privilege** reduces unauthorized access.

Q19. [BL5]

A cloud data center has frequent VM performance degradation. Monitoring shows high CPU utilization, high memory usage, and high storage latency. The organization has a limited budget. Which action should be prioritized first?

A. Create additional VMs on the same overloaded host
B. Analyze workloads and redistribute or migrate VMs to reduce resource contention
C. Increase the number of virtual networks
D. Disable all security mechanisms

Answer: B

Explanation: The immediate issue is **resource contention**. Workload redistribution or VM migration can reduce pressure without immediately requiring expensive infrastructure upgrades. Further capacity upgrades can be considered after monitoring.

Q20. [BL5]

A cloud provider wants to host finance, healthcare, and student workloads on shared physical infrastructure while maintaining strong isolation and efficient resource utilization. Which solution is the most appropriate?

A. Run all workloads directly on one operating system without virtualization

B. Use virtualization with hypervisor isolation, virtual networks, segmentation, and centralized SDDC management
C. Give every VM unrestricted access to every other VM
D. Disable network security to improve performance

Answer: B

Explanation: Virtualization enables resource isolation and consolidation, while **virtual networking and segmentation** control communication. SDDC management adds centralized control and automation. This provides a balanced approach to **security, resource utilization, and manageability**.